From Reviews to Retail Actions

NLP-Based Analysis of Drivers of Product Recommendation in Women's E-Commerce

NLP & Text Analytics

Term 5, 2nd Year, MBA

GITAM School of Business, GITAM Hyderabad

D.V Sri Nitya Nikhila- 2025011972

G. Madhusri- 2025228772

Rashmitha- 2025009327

Raghunandan- 2025360369

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Abstract

Customer reviews contain detailed descriptions of product experiences that are not fully captured by star ratings alone. This project applies Natural Language Processing (NLP) and supervised machine learning to women's e-commerce reviews to examine the language associated with product recommendation and to translate recurring patterns into practical retail insights.

The cleaned analysis dataset contains 22,640 reviews across 10 columns. The average customer rating is 4.18 out of 5, and 81.89% of reviews are marked as recommending the product. The analysis combines descriptive exploration, word and bigram frequency analysis, VADER sentiment scoring, TF-IDF feature engineering, and a Logistic Regression classifier. The text representation uses up to 5,000 unigram and bigram features, with an 80:20 stratified train-test split.

On the held-out test set, Logistic Regression achieved 86.95% accuracy, 95.93% precision, 87.78% recall, 91.68% F1-score, and 92.86% ROC-AUC. Compared with a majority-class Dummy Classifier, the model improved accuracy and F1-score. Terms such as "love," "perfect," and "comfortable" were associated with recommendations, while "disappointed," "returning," "returned," and "not flattering" were associated with non-recommendations. Frequent phrases also highlighted fit and sizing, including "true size," "fit perfectly," and "fit well."

The findings suggest that review analytics can help retail teams monitor fit, sizing, comfort, product descriptions, and recurring dissatisfaction. These are associations in one dataset, not proof of cause and effect. The model should be used as a decision-support aid alongside human review, particularly because recommendation labels are imbalanced and the data represents one retail context.

1. Introduction

1.1 Introduction to the Study

E-commerce has made customer reviews an important source of information for shoppers and retailers. A rating provides a compact evaluation, but the written review can explain the experience behind it: whether a garment fit as expected, whether the fabric felt comfortable, whether the appearance matched the product listing, or whether the customer encountered a problem.

The volume of review text makes manual reading difficult at scale. NLP offers a structured way to process text, identify recurring words and phrases, estimate sentiment, and build models that connect language to an outcome of interest. In this project, the outcome is the customer's product recommendation indicator.

The study follows a practical analytics sequence: describe the dataset, explore the language customers use, convert the text into numerical features, train and evaluate a classifier, and interpret the patterns in business terms. The goal is not to automate every retail decision, but to demonstrate how review text can become a useful source of evidence for further investigation.

1.2 Business Problem Definition

Retail teams may receive thousands of reviews across product categories yet lack a consistent method for identifying recurring language patterns that align with recommendation outcomes. Ratings can show that customers are dissatisfied, but they do not always explain whether the concern relates to fit, fabric, quality, appearance, or an expectation mismatch.

The business problem addressed here is: How can customer review text be analysed to identify language patterns associated with recommending or not recommending a product, and how can those patterns inform product information and customer-experience decisions?

The analytical task is a binary classification problem: predict the recommendation label from review text. The business interpretation task is broader: identify themes that deserve attention, while avoiding claims that a particular word or phrase directly causes a recommendation.

1.3 Research Objectives

- Describe the dataset using review counts, rating distribution, recommendation status, and selected department-level rates.
- Explore customer language through frequent words, two-word phrases, and sentiment categories.
- Convert review text into numerical TF-IDF features suitable for supervised classification.
- Train a Logistic Regression model to classify recommendation outcomes from review text.
- Evaluate the model using multiple metrics and compare its performance with a majority-class baseline.

- Interpret model-associated terms and descriptive patterns to develop practical, appropriately qualified business recommendations.

1.4 Scope of the Study

The study focuses on the cleaned Women's Clothing E-Commerce Reviews dataset and the recommendation label provided in that dataset. Descriptive analysis includes ratings, recommendation outcomes, selected product departments, and review text. The modelling scope is limited to the reported TF-IDF and Logistic Regression pipeline and its baseline comparison. The project does not establish causal drivers of customer behaviours, measure financial impact, or validate the model in a live retail environment.

2. Literature Review

2.1 Literature Review and Theoretical Background

Sentiment analysis and opinion mining are established NLP areas concerned with identifying evaluative language and opinions in text. Pang and Lee (2008) review methods for extracting subjective information, while Liu (2012) discusses opinion mining and sentiment analysis, including the importance of identifying opinions and the aspects to which they refer. These ideas are relevant to customer reviews because a review can contain both an overall judgment and comments about product attributes.

A central distinction in review analytics is between the polarity of a statement and the topic it discusses. For example, a review may mention "fit" positively ("fits perfectly") or negatively ("fits too small"). Word counts can identify what customers discuss, but interpretation requires context. This is why the project combines frequency analysis with sentiment scoring and a supervised classifier rather than relying on a single measure.

2.2 Existing Research on NLP in E-Commerce

In online retail, review analytics can support the organization of customer feedback, comparison of recurring product concerns, and summarization of language across products or categories. Lexicon-based sentiment tools assign polarity using predefined word and linguistic cues. VADER, introduced by Hutto and Gilbert (2014), is a rule-based sentiment model designed to account for sentiment-bearing lexical and stylistic signals. Its output is useful for broad exploratory analysis, although it can misread domain-specific language or context.

Supervised text classification approaches learn associations between labelled examples and text features. TF-IDF represents documents through weighted term frequencies, reducing the influence of terms that occur in many documents while emphasizing terms that are more distinctive. Linear models such as Logistic Regression are commonly used with sparse text features and have interpretable coefficients that can help summarize which terms align with each class.

2.3 Research Gap and Study Rationale

The practical challenge is not merely to label reviews as positive or negative; it is to connect patterns in review language to a business question that teams can investigate. This project

brings together descriptive text exploration, VADER sentiment labels, and a recommendation classifier in one workflow. It then interprets the results in relation to fit, sizing, comfort, product descriptions, and dissatisfaction-related language.

The study is exploratory and applied. It does not claim to introduce a new NLP algorithm. Its contribution is a compact end-to-end demonstration of how existing text-analysis methods can be applied to a retail review dataset and translated into cautious, actionable questions for business teams.

3. Data Collection and Preprocessing

3.1 Data Collection and Dataset Description

The project uses the Women's Clothing E-Commerce Reviews dataset. Following cleaning and preparation, the analysis dataset contains 22,640 records and 10 columns. Review text is the principal input for NLP, while the recommendation indicator is the target variable for classification. Rating and product department are used for descriptive analysis.

Dataset attribute	Description
Unit of analysis	One customer review record
Text input	Review text used for text exploration and model features
Target variable	Product recommendation indicator (recommended / not recommended)
Descriptive variables	Customer rating and product department
Prepared dataset size	22,640 records × 10 columns
Average rating	4.18 out of 5
Recommendation rate	81.89%

The dataset is dominated by positive outcomes: the recommendation rate is 81.89%. This imbalance is important for interpreting model performance, since a classifier can achieve high accuracy by predicting the majority class frequently.

3.2 Data Cleaning and Preprocessing

The data preparation stage produced a cleaned analysis table and a text field suitable for feature extraction. In general, preprocessing for review analytics must balance consistency with preservation of meaning. Standardizing text can make tokens easier to compare, but removing words indiscriminately can erase important information.

Negation is especially important in product reviews. Phrases such as "not flattering" express a different evaluation from "flattering." Similarly, return-related expressions may signal a customer experience that deserves attention. Therefore, preprocessing decisions should preserve meaningful terms and short expressions where possible, particularly when unigram and bigram features are used.

The available project outputs establish the final dataset size and model preparation settings, but do not provide a complete row-by-row cleaning log in this report. Any final submission should ensure that the notebook documents the exact handling of missing values, duplicate records, and text transformations used.

3.3 Data Preparation for Analysis

The recommendation indicator was used as the target label. Review text was transformed into a TF-IDF matrix with a maximum of 5,000 features, using unigrams and bigrams. The vectorizer settings were min_df=5 and max_df=0.95. These settings limit the feature vocabulary to terms that occur in a minimum number of documents and exclude terms that appear in an extremely large share of documents.

The dataset was split into training and testing partitions using an 80:20 stratified split. The training partition contains 18,112 records and the test partition contains 4,528 records. Stratification preserves the approximate target-class proportions across the two partitions, making the held-out evaluation more representative of the prepared dataset's class distribution.

Preparation component	Setting / value	Purpose
Feature extraction	TF-IDF	Represent review text numerically
Vocabulary size	Up to 5,000 features	Limit feature dimensionality
N-gram range	(1, 2)	Capture individual terms and two-word phrases
Minimum document frequency	5	Exclude extremely rare terms
Maximum document frequency	0.95	Exclude terms present in nearly all documents
Train-test split	80:20, stratified	Evaluate on held-out data while preserving class proportions
Training records	18,112	Fit the model
Testing records	4,528	Evaluate model performance

4. Exploratory Text Analysis and Feature Engineering

4.1 Exploratory Data Analysis

The initial exploration shows that customer feedback is concentrated in high ratings and positive recommendation outcomes. The average rating is 4.18 out of 5, and the 5-star category is the most common. Recommended reviews form the majority class, creating an imbalance that must be considered in model evaluation.

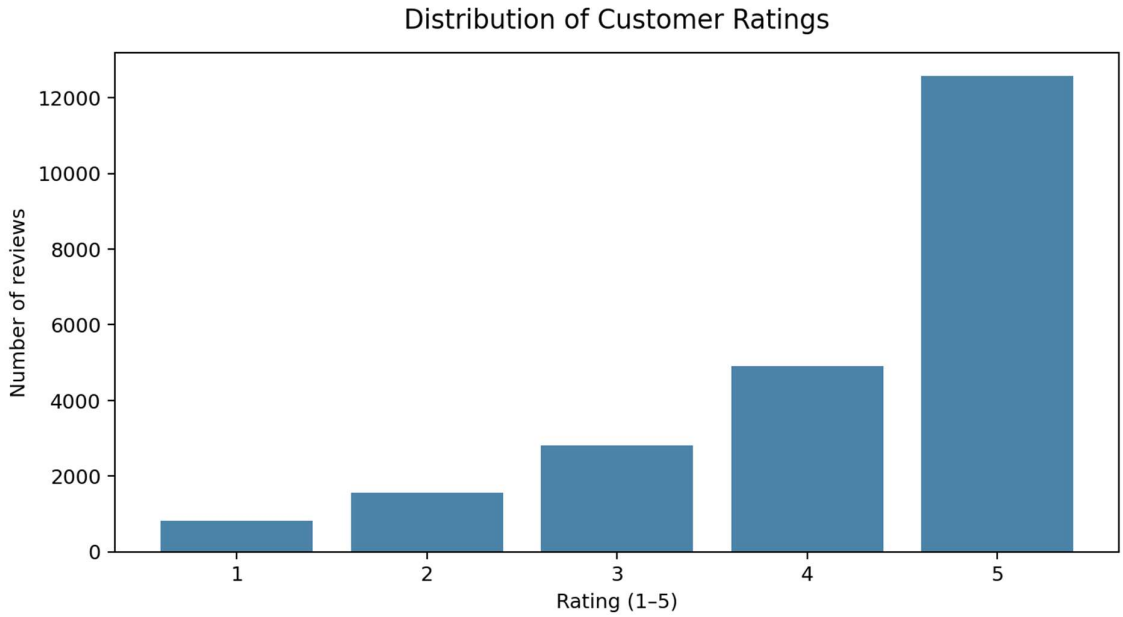


Figure 1. Distribution of customer ratings.

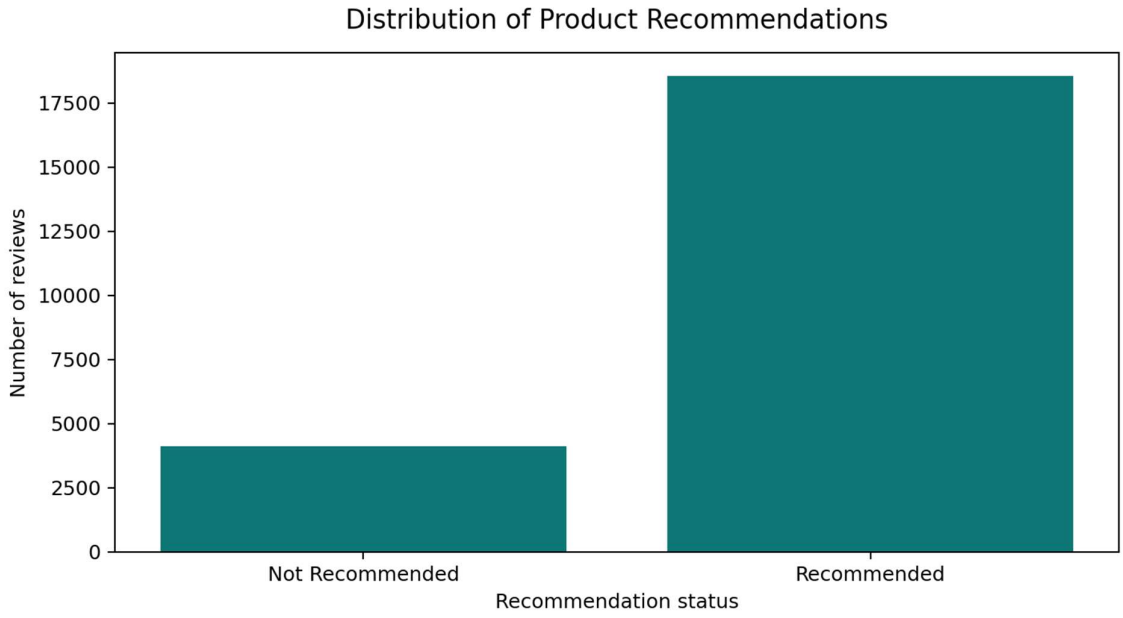


Figure 2. Distribution of product recommendation outcomes.

Department-level recommendation rates provide another descriptive view of the data. In the displayed summary, Bottoms has the highest rate at 84.95%, followed by Intimate at 84.63% and Jackets at 83.33%. Trend has the lowest displayed rate at 74.58%. These differences identify areas for further investigation; they do not explain why customers differ across departments.

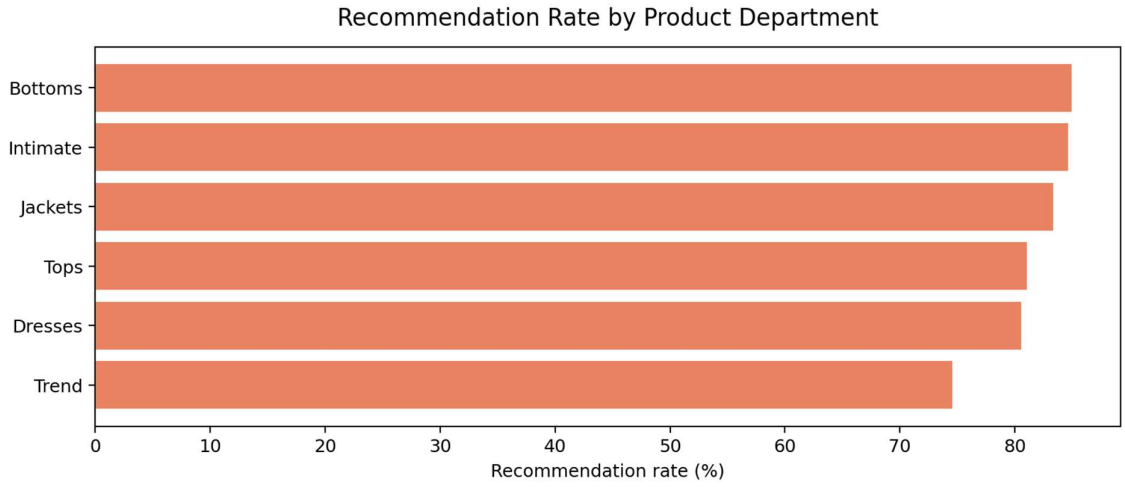


Figure 3. Recommendation rate by selected product department.

4.2 Exploratory Text Analysis

The review vocabulary frequently refers to product type and appearance, including “dress,” “top,” “color,” and “look.” Fit and sizing are also prominent, with terms such as “fit,” “size,” and “small.” This combination suggests that customers use reviews to describe both how an item looks and how it fits or feels when worn.

Frequency analysis is descriptive rather than evaluative. A frequent word may occur in praise, criticism, or a neutral description. For example, “size” can appear in a positive comment about true-to-size fit or a negative comment about an item running small. The frequency results are therefore interpreted alongside phrase-level patterns and sentiment/model outputs.

4.3 Word Frequency and N-gram Analysis

Word-frequency analysis identifies the most common individual tokens in the review text. The leading terms include “dress,” “fit,” “not,” “size,” “love,” “top,” “like,” “color,” and “look.” The presence of product-type, fit, and appearance vocabulary highlights the dimensions customers repeatedly discuss.

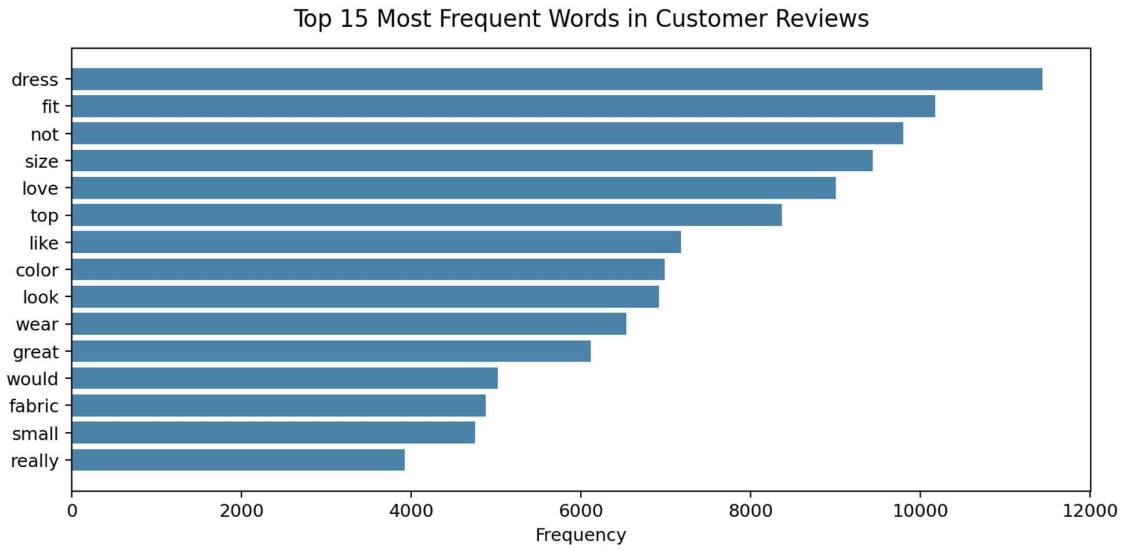


Figure 4. Top 15 most frequent words in customer reviews.

Bigram analysis examines consecutive two-word phrases, which can preserve a little more context than isolated tokens. Leading phrases include “true size,” “fit perfectly,” “look great,” “look like,” “love dress,” and “fit well.” These phrases show how fit and appearance are expressed in short combinations. Other phrases, such as “run large” and “size small,” point to potential sizing concerns.

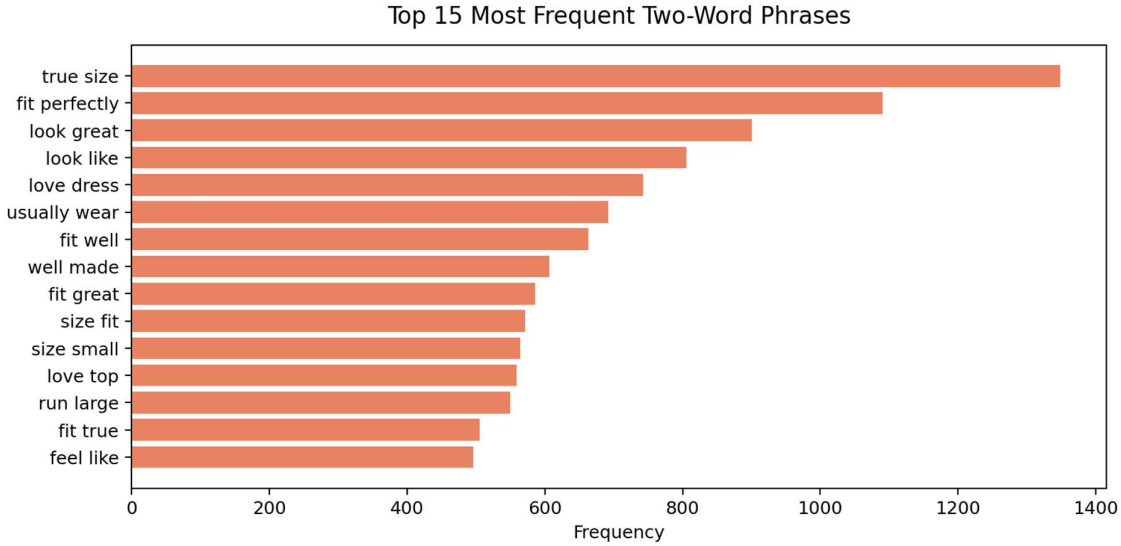


Figure 5. Top 15 most frequent two-word phrases in customer reviews.

Taken together, the frequency and bigram results suggest that fit, sizing, product appearance, and garment type are recurring review themes. They are useful for deciding what to inspect in more detail, but they do not by themselves establish customer satisfaction or the reasons behind recommendation decisions.

4.4 VADER Sentiment Analysis

VADER was used to assign a sentiment category to each review. The resulting distribution contains 20,986 positive reviews, 277 neutral reviews, and 1,377 negative reviews. Positive

labels account for the large majority of the dataset, while neutral reviews form a relatively small group.

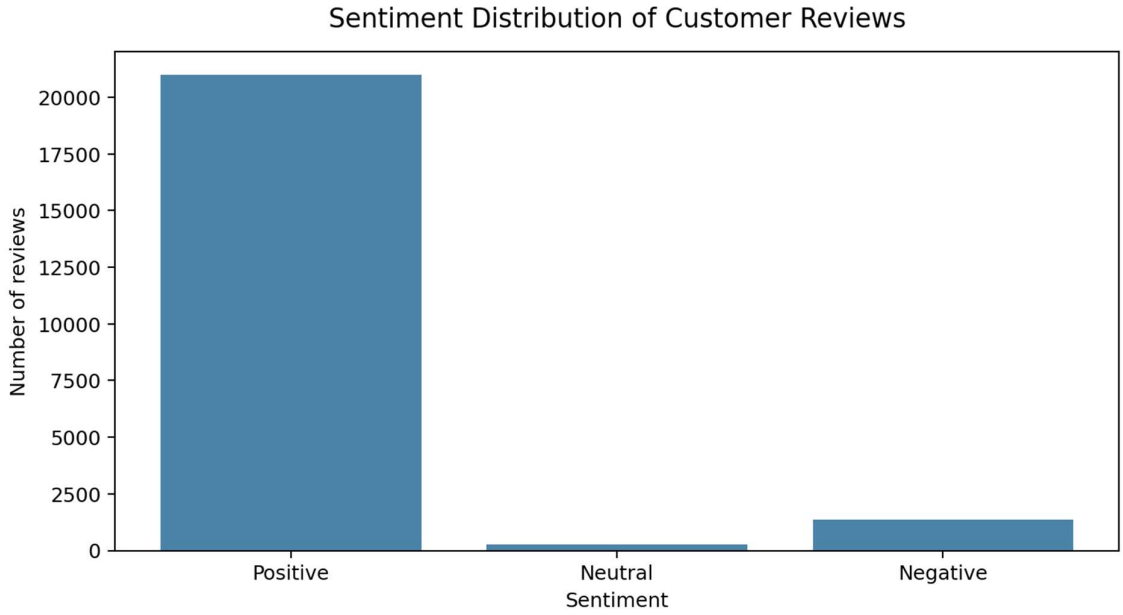


Figure 6. Distribution of VADER sentiment labels.

Recommendation rates differ across the sentiment categories: 85.43% for positive reviews, 47.29% for neutral reviews, and 34.86% for negative reviews. This pattern is consistent with an association between sentiment and recommendation, but the two measures are not interchangeable. The recommendation indicator is a dataset label, whereas VADER is a rule-based estimate derived from text.

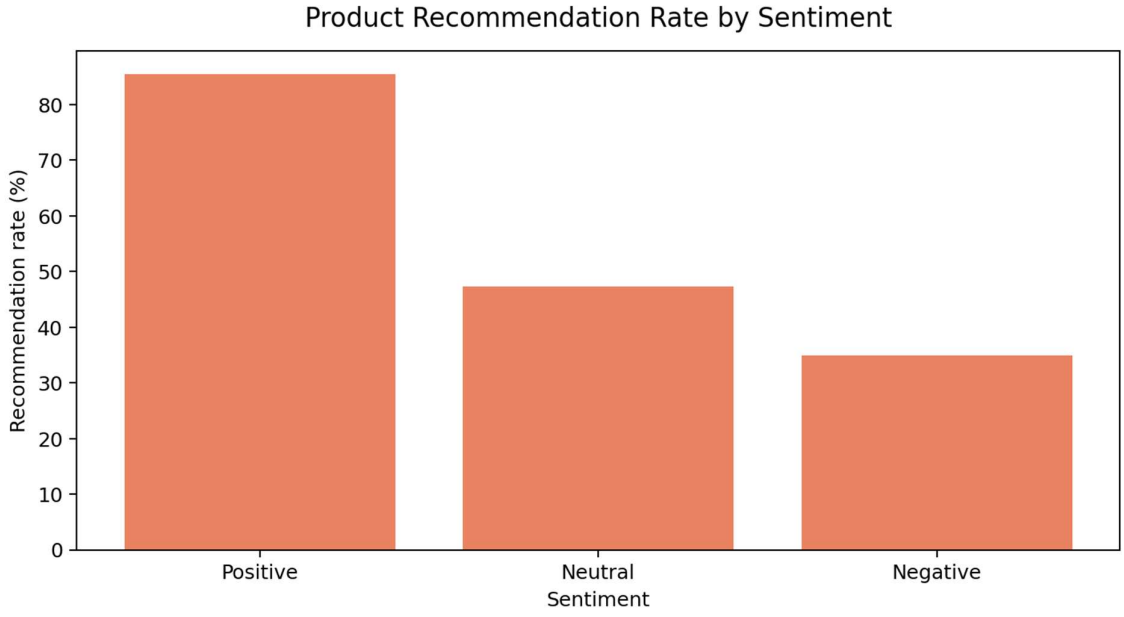


Figure 7. Product recommendation rate by VADER sentiment category.

The sentiment results should be treated as exploratory. VADER can provide a useful high-level signal, but its labels may not fully capture sarcasm, mixed opinions, product-specific

terminology, or nuanced descriptions. The small neutral category also means that estimates for neutral reviews may be less stable than those for the larger groups.

4.5 TF-IDF Feature Engineering

TF-IDF (term frequency–inverse document frequency) transforms text into numerical features. Term frequency reflects how often a term appears in a review, while inverse document frequency down-weights terms that appear across many documents. The result gives greater relative weight to terms that are more distinctive within the corpus.

The project used a vocabulary of up to 5,000 features and included both unigrams and bigrams. Unigrams allow the model to learn from individual words, while bigrams preserve short expressions such as “not flattering” or “true size.” The resulting sparse feature matrix was used as input to Logistic Regression.

TF-IDF is a representation of language, not a semantic understanding of the review. It does not automatically resolve ambiguity or determine whether a term is positive in every context. Model coefficients should therefore be read as learned associations within this dataset and feature setup.

5. Model Development and Evaluation

5.1 Model Selection and Development

The project uses Logistic Regression for binary classification. Despite its name, Logistic Regression is a classification method that estimates the probability of a class from a weighted combination of input features. With TF-IDF text features, the model learns which terms and phrases are associated with each recommendation outcome.

The model was fitted using the training partition and evaluated on the held-out test partition. A majority-class Dummy Classifier was used as a baseline. The baseline provides a simple reference point: it shows how well a trivial model performs by always predicting the most common class, without learning patterns from the review text.

5.2 Logistic Regression

Logistic Regression is suitable for this project because it can handle high-dimensional sparse text features and produces coefficients that can be inspected. Positive coefficients indicate terms associated with the recommended class under the model's class coding; negative coefficients indicate terms associated with the non-recommended class. The magnitude of a coefficient reflects its contribution within the fitted linear model, not a causal effect.

The classification target is recommendation status. The model's predictions can help surface review language associated with the two outcomes, but an operational system would still need threshold selection, monitoring, error review, and validation on new data before being used in a business process.

5.3 Evaluation Metrics

Metric	What it measures	Why it matters here
Accuracy	Share of all predictions that are correct	Useful overall, but can look high when one class dominates
Precision	Share of predicted positive cases that are truly positive	Indicates how reliable recommended-class predictions are
Recall	Share of actual positive cases correctly identified	Shows how many recommended reviews the model captures
F1-score	Harmonic mean of precision and recall	Balances precision and recall in one measure
ROC-AUC	Ability to rank positive cases above negative cases across thresholds	Summarizes discrimination beyond one fixed classification threshold

Because the dataset is imbalanced, accuracy alone is insufficient. Precision, recall, F1-score, ROC-AUC, and the confusion matrix provide a more complete picture of performance. The interpretation of precision and recall depends on which class is designated as positive; the reported classification metrics here correspond to the recommended class.

5.4 Evaluation and Confusion Matrix

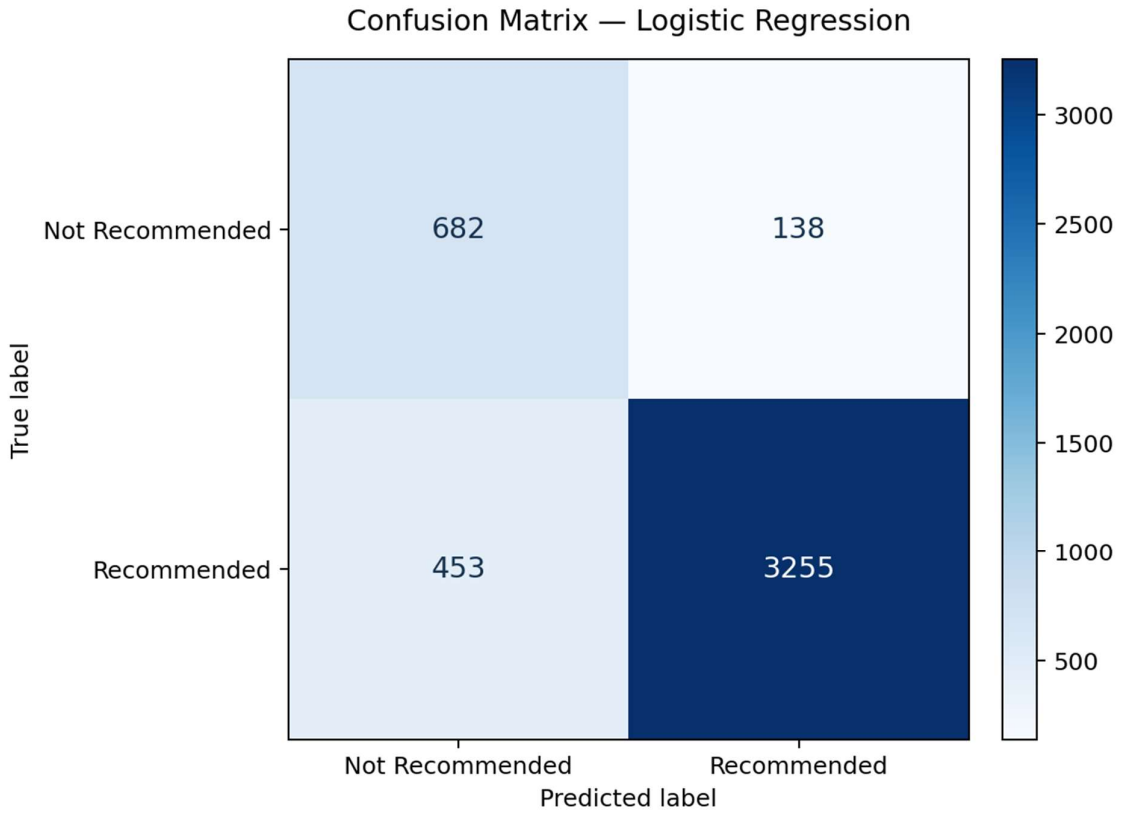


Figure 8. Confusion matrix for Logistic Regression on the held-out test set.

The confusion matrix contains 682 true non-recommendations correctly predicted as non-recommended and 3,255 recommendations correctly predicted as recommended. It also shows 138 non-recommended reviews incorrectly predicted as recommended (false positives) and 453 recommended reviews incorrectly predicted as non-recommended (false negatives).

The model identifies a large share of recommended reviews, while the false-negative count indicates that some genuinely recommended reviews are missed. The false-positive count represents cases where a review labeled as not recommended is classified as recommended. Which error matters more depends on the intended business use: for example, a system that flags potentially dissatisfied customers may need a different operating threshold from one that summarizes broad review trends.

5.5 Baseline Comparison

The Dummy Classifier achieved 81.89% accuracy and 90.04% F1-score. Logistic Regression achieved 86.95% accuracy and 91.68% F1-score. The model therefore improves on the baseline in both reported measures, suggesting that review text contains predictive information beyond simply assigning every record to the majority class.

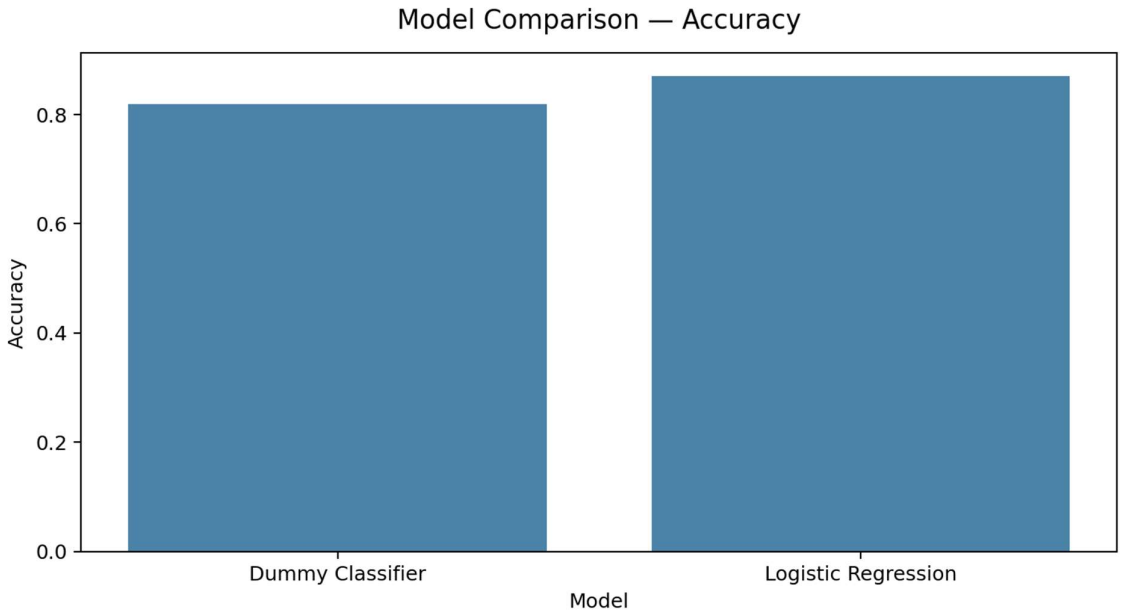


Figure 9. Accuracy comparison: majority-class Dummy Classifier and Logistic Regression.

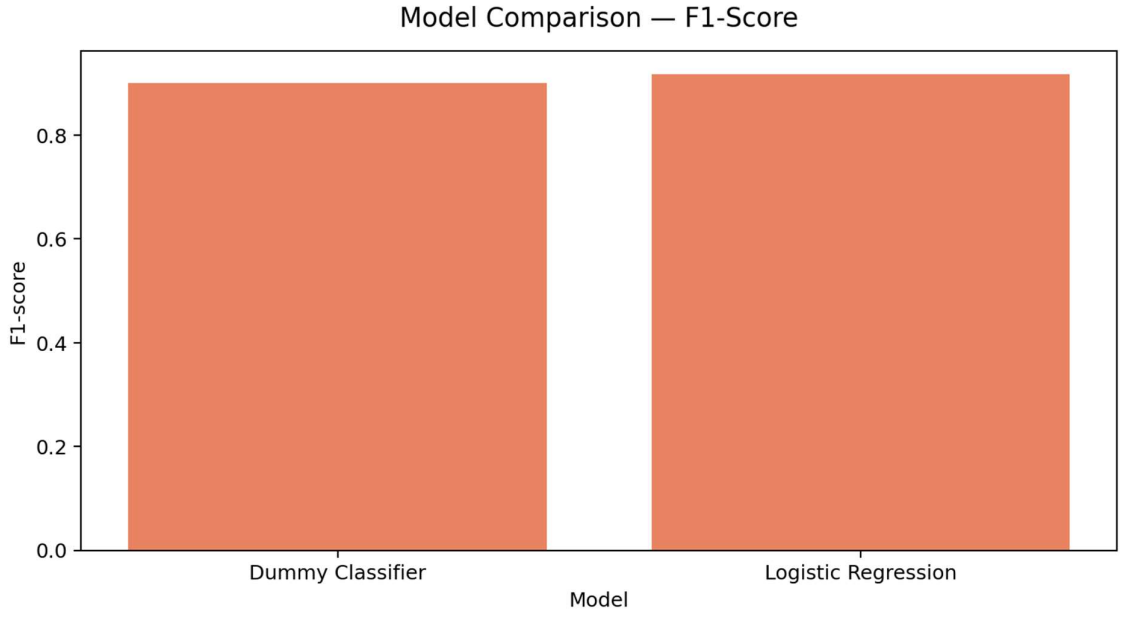


Figure 10. F1-score comparison: majority-class Dummy Classifier and Logistic Regression.

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Majority-class Dummy Classifier	81.89%	—	—	90.04%	—
Logistic Regression	86.95%	95.93%	87.78%	91.68%	92.86%

The baseline comparison should not be interpreted as proof that the classifier is ready for deployment. It is a single held-out evaluation, and the majority class makes accuracy appear

strong even for a model that does not learn review language. Additional validation, class-specific reporting, and error analysis would help establish how reliably the model performs.

5.6 Feature Importance and Coefficient Interpretation

The strongest positive coefficients include terms such as “love,” “perfect,” “comfortable,” and “great.” These expressions are associated with the recommended class in the fitted model and broadly align with language of satisfaction, comfort, and favorable product experience.

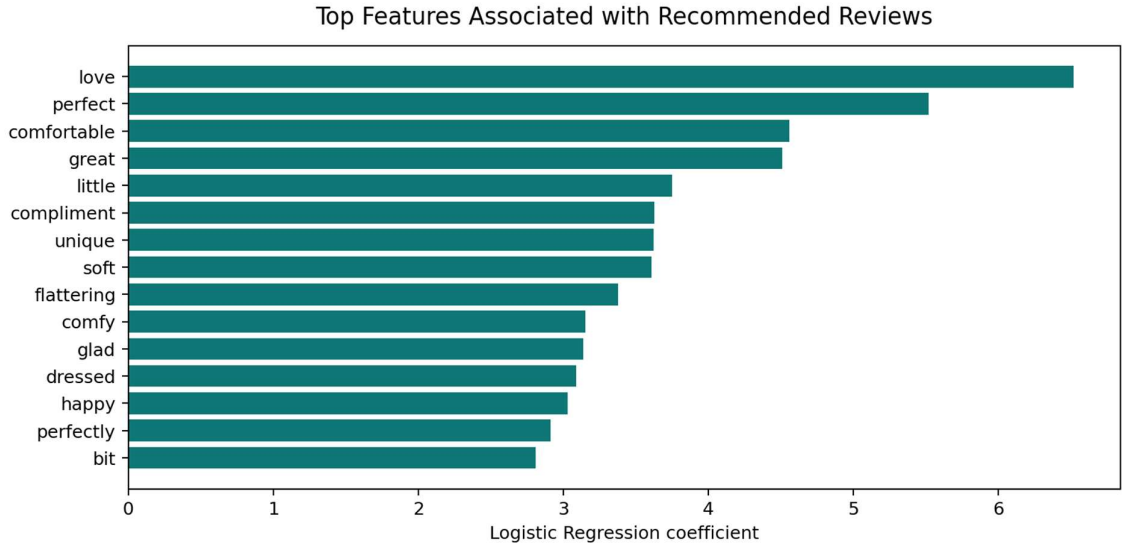


Figure 11. Terms with the largest positive Logistic Regression coefficients.

The strongest negative coefficients include “disappointed,” “wanted love,” “returning,” “returned,” and “not flattering.” These expressions are associated with the non-recommended class and point toward disappointment, returns, and mismatch between product expectations and experience.

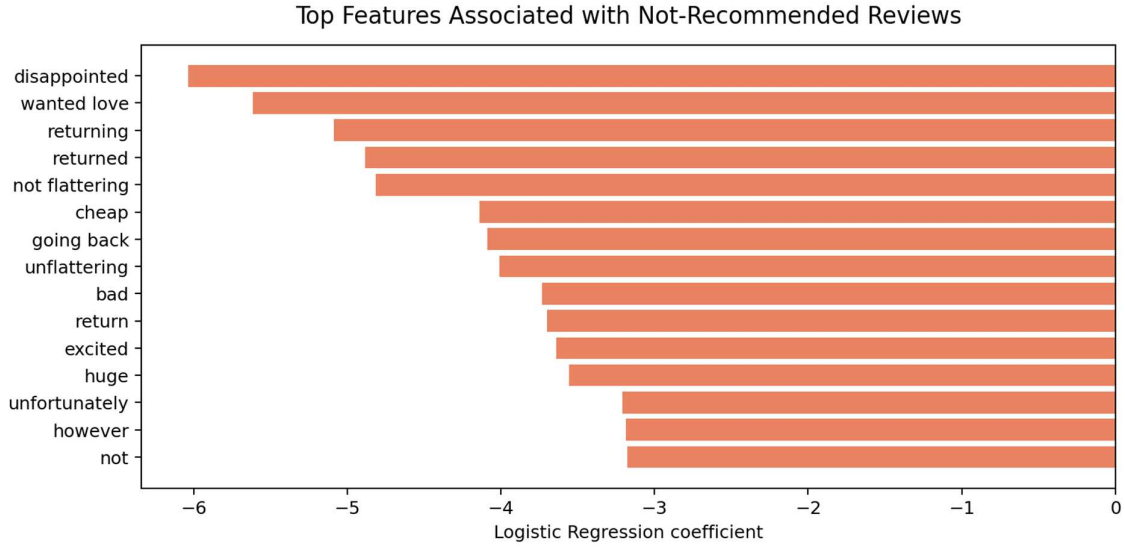


Figure 12. Terms with the largest negative Logistic Regression coefficients.

Coefficient interpretation has important limits. A term may be correlated with other words, and the model does not infer the underlying reason a customer wrote a review. Coefficients identify predictive associations in this sample; they should be used to guide qualitative review, not to claim that a phrase independently determines recommendation.

6. Results and Visualization

6.1 Dataset-Level Findings

- The prepared dataset contains 22,640 reviews, with an average rating of 4.18 out of 5.
- The recommendation rate is 81.89%, indicating a substantial majority class.
- The rating distribution is concentrated toward higher scores, with 5-star reviews most common.
- Displayed department recommendation rates range from 74.58% for Trend to 84.95% for Bottoms.

6.2 Sentiment and Text Findings

- VADER classified 20,986 reviews as positive, 277 as neutral, and 1,377 as negative.
- Positive reviews had an 85.43% recommendation rate; neutral reviews had 47.29%; negative reviews had 34.86%.
- Frequent terms point to product type, appearance, fit, and sizing as recurring discussion topics.
- Frequent phrases such as “true size,” “fit perfectly,” and “fit well” highlight sizing and fit language.
- Model-associated positive and negative terms reflect satisfaction, comfort, disappointment, and return-related experiences.

6.3 Model Performance Findings

- Logistic Regression achieved 86.95% accuracy, 95.93% precision, 87.78% recall, 91.68% F1-score, and 92.86% ROC-AUC.
- The model improved accuracy and F1-score compared with the majority-class Dummy Classifier.
- The confusion matrix shows both false positives and false negatives, reinforcing the need to review errors before practical use.

6.4 Overall Interpretation

Across the descriptive, sentiment, and predictive analyses, review language contains useful signals associated with recommendation status. Fit, sizing, comfort, appearance, and dissatisfaction-related expressions are prominent across the outputs. The results provide a basis for targeted review monitoring and product-information improvements, while leaving open questions about causality, generalizability, and performance on future reviews.

Evidence source	Observed pattern	Possible business interpretation
Rating and recommendation summaries	High average rating and majority recommendation	Customer feedback is generally favorable in this dataset, so negative signals may be less frequent

Word and bigram frequencies	Fit, size, appearance, and product-type language recur	These topics are candidates for structured review monitoring
VADER sentiment	Positive sentiment has a higher recommendation rate than negative sentiment	Sentiment can provide a broad triage signal, but not a substitute for the recommendation label
Model coefficients	Positive and negative language clusters align with satisfaction and dissatisfaction	Terms can help surface reviews or themes for human inspection
Department rates	Recommendation rates differ across selected departments	Category-level comparison may reveal where further review is useful

7. Business Insights and Recommendations

7.1 Business Insights

Fit and sizing are recurring themes

Terms and phrases such as “fit,” “size,” “true size,” “fit well,” “run large,” and “size small” appear frequently. This suggests that customers repeatedly use reviews to explain how a garment fits relative to their expectations. The data does not establish that sizing is the sole or primary cause of non-recommendation, but it supports examining fit-related feedback as a distinct theme.

Comfort and overall product experience matter in positive language

Words such as “comfortable,” “soft,” “perfect,” and “love” are associated with recommended reviews in the model. These terms may reflect favorable experiences with wearability, feel, or overall satisfaction. Retail teams can use them as examples of language customers use when describing a positive experience.

Dissatisfaction and return-related language can help identify review themes

Terms such as “disappointed,” “returned,” “returning,” “cheap,” and “not flattering” are associated with non-recommendation. These expressions may point to a mismatch between expectations and experience, dissatisfaction with appearance or perceived quality, or a return-related issue. Each theme should be verified by reading representative reviews before drawing product-level conclusions.

Department-level rates invite more granular investigation

The displayed recommendation rates vary across departments. A lower department rate can be a useful signal for deeper analysis, but the comparison does not account for factors such as product mix, review volume, price, or customer composition. Department-level results should therefore be treated as a starting point rather than a performance ranking.

7.2 Recommendations

1. Improve product descriptions around fit and sizing

Review teams can identify recurring size-related expressions and compare them with product-page information. Where reviews repeatedly mention an item running large or small, the business could review size charts, fit notes, garment measurements, and model-fit descriptions. Any change should be validated against product-specific review evidence rather than applied uniformly across a category.

2. Make fabric, feel, and appearance details more explicit

Positive and negative review language frequently touches on comfort, fabric, look, and expectations. Product pages can be reviewed for clarity about material composition, stretch, texture, thickness, drape, and appearance. The purpose is to reduce avoidable expectation gaps by communicating product characteristics clearly and consistently.

3. Establish a recurring review-monitoring workflow

A practical workflow could summarize review volume, sentiment distribution, common phrases, and recommendation rates at a regular cadence. A sudden increase in negative or return-related language could trigger a human review of the underlying comments. The model should flag patterns for investigation rather than automatically determine product quality or customer intent.

4. Analyze feedback at product and category level

Department-level averages can hide variation among individual products. Where sufficient review volume exists, teams can compare fit-related and dissatisfaction-related themes by product, department, and time period. Comparisons should include sample sizes and avoid over-interpreting small groups.

5. Route recurring themes to the relevant teams

Review themes can be organized into operational categories—such as sizing, comfort, appearance, perceived quality, and returns—and shared with the teams best positioned to investigate them. Product development may examine recurring design or fit concerns; merchandising may refine product descriptions; customer experience teams may use themes to improve support and review response practices.

6. Keep a human-in-the-loop review process

Before acting, teams should inspect representative positive and negative reviews, confirm that a detected phrase is being interpreted correctly, and check whether the pattern is concentrated in a small number of products. Human review helps reduce the risk of acting on sarcasm, ambiguous wording, or misleading model predictions.

7.3 Practical Applications

Potential application	How the analysis could be used	Safeguard
Review dashboard	Track review volume, recommendation rate, sentiment labels, and recurring terms	Show sample sizes and sentiment/model limitations
Fit-feedback monitor	Surface terms and phrases related to sizing and fit	Validate phrases against actual review text and product details
Product-page review	Identify recurring gaps between stated product features and customer descriptions	Use product-specific evidence before editing descriptions
Customer-experience triage	Flag clusters of negative or return-related language for review	Do not automatically infer customer intent or trigger actions solely from a score
Category-level insight report	Compare themes across departments and time periods	Account for differences in volume and product mix

8. Conclusion and Future Scope

8.1 Conclusion

This project demonstrates an end-to-end NLP workflow for analysing women's e-commerce reviews and predicting product recommendation status. The analysis combined descriptive exploration, word and bigram frequencies, VADER sentiment scoring, TF-IDF feature engineering, and Logistic Regression. The prepared dataset contains 22,640 reviews, has an average rating of 4.18 out of 5, and an 81.89% recommendation rate.

The text analysis repeatedly highlighted fit, sizing, product appearance, comfort, and garment-related vocabulary. VADER sentiment labels showed a higher recommendation rate among positive reviews than among neutral or negative reviews. Logistic Regression achieved 86.95% accuracy and a 91.68% F1-score, exceeding the majority-class baseline on the reported accuracy and F1 measures.

The practical value of the project lies in identifying themes that retail teams can investigate, not in claiming that the model has discovered causal drivers of recommendation. The results support a cautious review-analytics workflow in which automated signals are combined with human interpretation and additional validation.

8.2 Limitations

- Class imbalance: recommended reviews from the majority class, which can inflate the apparent usefulness of accuracy and complicate performance on the minority class.
- Single-dataset scope: the dataset represents one retail context, so results may not generalize to other retailers, product categories, or customer populations.
- Sentiment limitations: VADER may miss sarcasm, mixed sentiment, context, and product-specific meaning; its labels are not human-validated ground truth in this project.
- Text representation limits: TF-IDF captures weighted terms and short phrases but does not fully represent semantic context or relationships across a review.
- Associational interpretation: model coefficients indicate predictive associations and do not prove that a word, product attribute, or experience causes recommendation behavior.
- Single split: reported metrics come from one train-test split and may vary with a different sample, preprocessing choice, or classification threshold.
- Limited operational validation: the project does not test live deployment, measure business impact, or establish an ongoing monitoring process.

8.3 Future Work

- Report class-specific precision, recall, and F1-score, and inspect performance separately for recommended and non-recommended reviews.
- Use cross-validation and repeated evaluation to assess whether model performance is stable across different data partitions.
- Conduct structured error analysis on false positives and false negatives, including representative review excerpts and possible causes of misclassification.
- Test class-weighting or decision-threshold adjustments if the intended use requires stronger detection of non-recommended reviews.

- Compare Logistic Regression with additional suitable text-classification approaches while retaining a clear baseline and consistent evaluation protocol.
- Explore aspect-based sentiment analysis to separate feedback about fit, fabric, comfort, quality, and appearance.
- Validate the approach on another dataset or a later time period to assess generalizability and changes in customer language.
- Build a prototype dashboard that presents trends, sample sizes, confidence or uncertainty indicators, and links to the underlying reviews for human verification.

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Annexures

Google Colab Notebook:



Code_NLP Project.pdf

Women's Clothing E-Commerce Reviews Dataset:



Womens Clothing
E-Commerce Reviews